Three-class Motor Imagery Classification Based on FBCSP Combined with Voting Mechanism

Bo Li, Banghua Yang*, Cuntai Guan, *Fellow, IEEE*, and Chenxiao Hu

Abstract— **Common Spatial Pattern (CSP) is an effective algorithm in constructing optimal spatial filters, which is widely used to discriminate two classes of electroencephalogram (EEG) signal in Motor Imagery (MI) based Brain Computer Interface (BCI). To extend CSP algorithm to three-class motor imagery of left-hand, right-hand, both-feet, in this paper a three-class classification strategy based on Filter Bank Common Spatial Pattern (FBCSP) and voting mechanism is proposed. The strategy reduces a three-class problem to two binary-class problems. Two binary-class classifiers are constructed for the three-class classification, both-hands vs both-feet and left-hand vs right-hand. The result shows an average three-class classification accuracy of 68.6% with BCI competition IV Datasets 2a, which is an encouraging result in motor imagery pattern recognition. And demonstrated that both-hands can be considered as one class in MI based BCI system.**

Keywords—**FBCSP; motor imagery; three-class classification; voting mechanism;**

I. INTRODUCTION

Motor imagery（MI）based Brain-computer interface (BCI) is playing an increasingly important role in post-stroke rehabilitation and other fields. The challenge in MI-based BCI (MI-BCI) systems, which translate the mental imagination of self movement to specific commands, is the huge inter-subject variability with respect to the characteristics of the brain signals [1].

Different parts of human body have a spatial mapping in the brain primary motor cortex. Hence, spatially discriminable patterns of EEG signals are theoretical basis of MI-BCI. The Common Spatial Pattern (CSP) method is widely used to construct optimal spatial filter, which has achieved good results in MI-BCI. In 1991, Z.J. Koles first proposed that CSP can be used to extract the discriminative features in EEG [2]. In 1999, Möller-Gerking. J successfully applied CSP to process motor imagery EEG signal for the first time [3] [4]. However, the original CSP algorithm has the limitations of too many EEG-leads needed and lack of frequency domain information. Hence, In 2008, Kai Keng Ang proposed the Filter Bank Common Spatial Pattern (FBCSP) to perform autonomous selection of key temporal-spatial discriminative EEG characteristics for CSP and achieved higher accuracies compared to other methods[5].

However, the algorithms above can only be used for binary-class motor imagery classification, because CSP itself is a spatial filtering algorithm for binary classification. To address the issue multi-class extensions to CSP is proposed, which reduces a multi-class to several binary problems is suggested in [5] for CSP in a BCI context. Three approaches of multi-class extension to the FBCSP algorithm present in [7], The results showed the One Versus-Rest (OVR) approach achieved best classification result(0.569) in four-class motor imagery.The OVR approach discriminates each class form the rest classes, 4 binary-class classifier should be prepared for four-class classification.

In this paper another multi-class motor imagery classification method based on FBCSP combined with voting mechanism is proposed . In this method, 2 binary-class classifiers should be constructed for the three classes of motor imagery. The training sets labeled as left-hand and right-hand motor imagery are merged as a new class (both-hands), the training sets of both-feet as the other class. Then modeling both-hands and both-feet EEG signal to obtain the first classifier. Then, the training sets labeled as left-hand and right-hand motor imagery are respectively taken out, the second classifier of right-hand and left-hand is constructed.

In the test process, test sets are input into the first classifier to obtain preliminary prediction classification result according to the voting mechanism. If the preliminary result is both-feet, the final classification prediction result is both-feet simultaneously. If the classification result is both-hands, the test sets will be send to the second classifier to obtain the final classification prediction result.

Bo Li is with the Department of Automation, Shanghai University of China (e-mail: libo_go@163.com).

Banghua Yang (Corresponding author) School of Mechatronic Engineering and Automation, Research Center of Brain Computer Engineering, Shanghai University, Shanghai 200072, China) (e-mail: yangbanghua@shu.edu.cn).

Cuntai Guan School of Computer Science and Engineering, Nanyang Technological University, 50 Nanyang Avenue, 639798, Singapore (e-mail: CTGuan@ntu.edu.sg).

Chenxiao Hu is with the Department of Automation, Shanghai University of China (e-mail: huchenxiao8973@163.com).

II. METHODS

A. Introduction to FBCSP

The CSP algorithm is used to extract the most discriminative features in motor imagery EEG signal. The principle basis is to use the diagonalization of the matrix to find a set of optimal spatial filters for projection, to maximize differences in variance of two types of signal, then the feature vector with higher discrimination is obtained. Assume that the raw EEG in a single motor imagery trial is E_{N*T} , N is the number of electrode channels, T is each channel number of Samples. The required spatial filter matrix $W=\left(B^T P\right)^T$ can be calculated，the specific calculation process is shown in [8]. By using CSP feature extraction, a feature vector $F=\left[f_1, f_2, ..., f_{2m}\right]$ can be finally obtained. m is the number of the feature pairs selected.

In the FBCSP algorithm, the EEG signal is firstly bandpass-filtered into multiple frequency bands, then CSP features are extracted from each band. Subsequently discriminative pairs of frequency bands and corresponding CSP features are selected automatically with specific feature selection algorithm. In this algorithm, the process of manually selecting a subject-specific frequency range in the original CSP algorithm is omitted.

B. Introduction to Voting Mechanism

Typically, the CSP algorithm usually performs feature extraction with a wide frequency band of 4-40 Hz, which contains most of the information related to motor imagery, but also contains some redundant information. Therefore, based on the core idea of FBCSP, wide frequency band of the EEG signal was decomposed into 18 sub-bands with a specific 2Hz bandwidth in this paper. Then CSP features are respectively extracted from each sub-bands. A SVM classification algorithm is subsequently used to classify the CSP features of each sub-bands. 18 classification results are obtained to vote with different weights for the final classification result. Mu rhythm (8-12Hz) and Beta rhythm (18-26Hz) are most relevant rhythm to motor imagery [9], corresponding voting weight is set to 1/10, higher than the other bands 1/30. The weighted sum of 18 results is calculated as the final classification result.

C. Three-class classification strategy

The approach includes two processes: modeling process and testing process. The specific modeling process is shown in Fig. 1, which is described as follows:

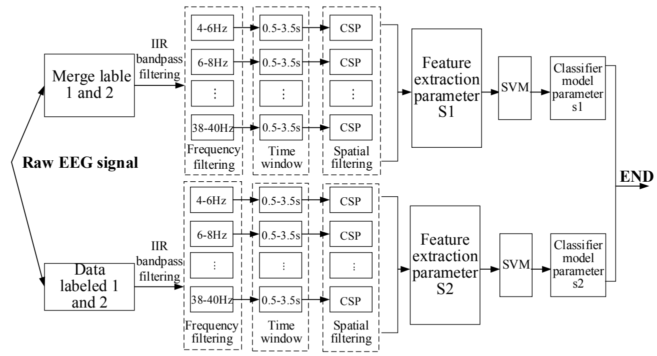

Fig. 1. Modeling Process.

(1) The training sets labeled as left-hand motor imagery (label 1) and right-hand motor imagery (label 2) are merged into one class (both-hands, set new label 0), and the training sets of both-feet (label 3) as the other class.

(2) In reference [5], the smaller the bandwidth is, the more sub-bands are decomposed, and the richer frequency information is provided. The EEG signal is decomposed from 4-40hz through a set of Infinite Impulse Response (IIR) bandpass filters, and decomposed into 4-6 Hz, 6-8 Hz..., 38-40Hz a total of 18 frequency bands.

(3) For each decomposed frequency band signal, use a time window to extract data from 0.5s to 3.5s， then feature extraction is carried out with CSP algorithm, 2-dimensional spatial filter and 2-dimensional training feature are obtained. In the step feature extraction model with a total of 36-dimensional is saved（with parameters $S1$ ）.

(4) Each frequency band 2-dimensional training feature and corresponding label are input into the SVM classifier, then the classifier parameter $s1$ ($1*1$ structure) is obtained. Finally, the classifier parameter $1*18$ structure is composed.

(5) Extract the data corresponding to the labels 1 and 2 in the raw EEG (remove the trial containing NaN), repeat the above steps (2) - (4), and obtain the second trained model with feature extraction model parameter $S2$ and the classifier model parameter $s2$.

The test process is shown in Fig. 2, which is described as follows:

(1) Select a test trial from the test sets. If it contains NaN, then reject it and reselect, if NaN is not included, perform the following steps.

(2) The selected data is decomposed into 18 frequency bands at a bandwidth of 2 Hz from 4-40 Hz by an IIR bandpass filter.

(3) Use a time window to extract 0.5-3.5s data from each frequency band signal and input it into the CSP filter with the parameter $S1$ from feature extraction model obtained in step 3 of the training process. Two-dimensional test feature is extracted to form a 36-dimensional test feature which is a set of feature vectors $F_1=\left[f_1^{'}, f_2^{'}, \cdots, f_{36}^{'}\right]$.

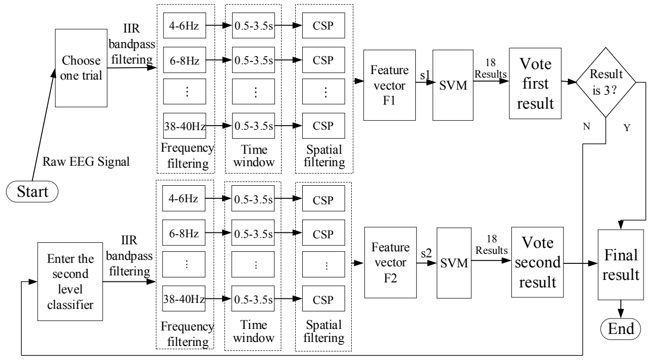

Fig. 2. Test Process

(4) Input the two-dimensional test feature and the classifier model parameter $s1$ obtained in step 3 of the training process into the SVM classifier one by one, then obtain 18 classification results.

(5) 18 results participate in the voting with different weights, the first classification prediction result is determined based on the voting result.

(6) If the first classification prediction result is 3, the final classification prediction result is 3, the test process stops; if the result is 0, perform the second classification by steps 7 and 8 of the test process.

(7) Repeat step (2) of the test process.

(8) Input each frequency band signal into the CSP filter with feature extraction model paramete $S2$. Two-dimensional test features are extracted to form a 36-dimensional test feature which is a set of feature vectors $F_2 = \left[f_1^{'}, f_2^{'}, \cdots, f_{36}^{'} \right]$.

(9) Input the two-dimensional test feature and the classifier model with parameter $s2$ into the SVM classifier one by one to obtain 18 classification results.

(10)18 results participate in the vote with different weights, and the final classification prediction result is determined based on the voting result.

(11) Repeat steps (1)- (10) of the test procedure. Then obtain the classification prediction results of all trials of the test sets. Compare the final classification prediction result with the original label. Then calculate the correct rate.

III. EXPERIMENTAL RESULTS

A. Dataset

The data set used in this paper was from the 'Graz data set A' provided by the BCI Competition IV held on July 3, 2008. The data set included nine subjects (IDs 1-9, respectively), and each subject data included four categories: left hand, right hand, both feet, and tongue motor imagery. The number of acquisition channels is 22 with sampling rate 250Hz, electrode Montage is shown in Fig. 3. The raw EEG signal is bandpass filtered by 0.5Hz-100Hz.

The paradigm is illustrated in Fig. 4, the subjects were sitting in a comfortable armchair in front of a computer screen. At the beginning of a trial (t=0s), a fixation cross appeared on the black screen. After two seconds (t = 2 s), an arrow pointing either to the left, right, down or up (corresponding to one of the four classes left hand, right hand, foot or tongue) appeared and stayed on the screen for 1.25 s. The subjects were ask to carry out the motor imagery task until the fixation cross disappeared from the screen at t=6s. A short break followed where the screen was black again. More details are described in [10].

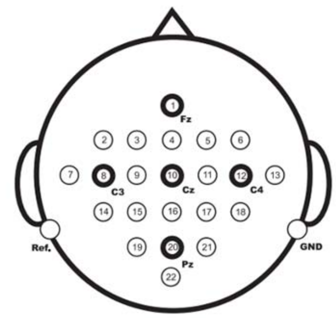

Fig. 3. Electrode montage corresponding to the international 10-20.

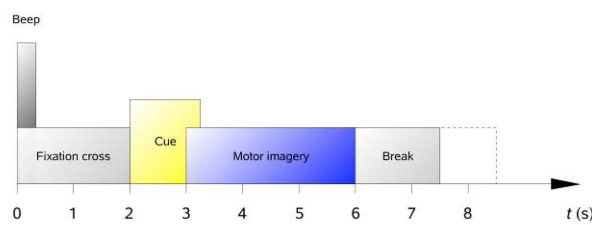

Fig. 4. Timing scheme of the paradigm.

In the competition data, each subjects' data comprised training sets and test sets. Both sets contain 72 left-hand motor imagery tasks, 72 right-hand motor imagery tasks, 72 both-feet motor imagery tasks, and 72 tongue motor imagery tasks. Firstly, the data of the tongue movement imagination task was removed, and the data of the three types of motor imagery tasks of the left hand, the right hand and the both-feet are obtained.

B. Preliminary Result

The training sets for each subject was all used for modeling and the test data was used for testing classification accuracy. The data is processed as described above, the classification accuracy of the nine subjects is shown in Table 1, the three-class classification accuracy is 68.6% averaged over the 9 subjects, the highest(subject ID=1) classification accuracy rate reached 84.11%.

In each trial, the subjects' motor imagery last 4s, but both the beginning and the end of motor imagery had the problem of thinking delay and visual evoked potential, which reducing the signal-to-noise ratio of motor imagery and thus reducing the accuracy of classification. To further study, five time windows of 0-3s, 1-4s, 0.5-3s, 1-3s and 0.5-3.5s were selected in the experiment, the result is shown in Table 2.

Table 1 Three-class motor imagery classification accuracy of 9 subjects, with the BCI Competition IV Data sets 2a

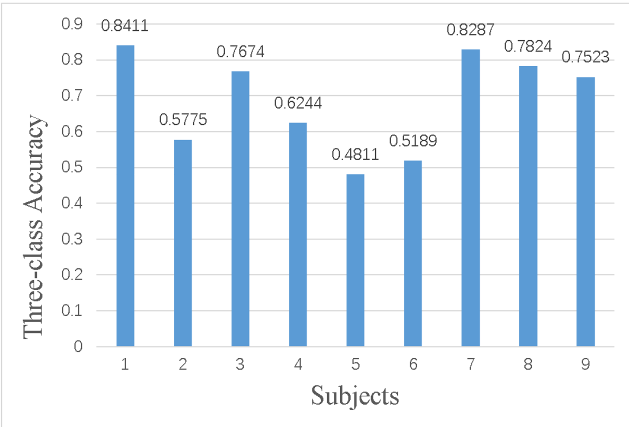

Table 2 Three-class motor imagery classification accuracy of 9 subjects with different time window, with the BCI Competition IV Data sets 2a

Window / subjects	[0s, 3s]	[1s, 4s]	[0.5s, 3s]	[1s, 3s]	[0.5s, 3.5s]
1	0.7804	0.7710	**0.8411**	0.729	0.8224
2	0.5668	**0.5936**	0.5775	0.5187	0.5668
3	0.7488	0.7023	0.7674	0.7302	**0.7907**
4	0.5587	**0.6291**	0.6244	0.5962	0.6244
5	0.4575	0.4434	0.4811	**0.5519**	0.4528
6	0.5566	0.4811	0.5189	0.4764	**0.5708**
7	0.7917	0.7824	**0.8287**	0.7361	0.8148
8	0.6991	0.6944	**0.7824**	0.7361	0.7546
9	0.7290	0.6729	**0.7523**	0.6916	0.7056
Ave	0.6543	0.6411	**0.6860**	0.6407	0.6781

The results showed that: (1) when time window 0.5-3s is selected, the classification accuracy is the highest, reaches to 68.6% averaged over 9 subjects, which verifies that the time window can reduce some of the negative impact of thinking delay and improve the classification accuracy. If each subject takes time window with the best result, the average accuracy rate of the 9 experimenters is 70.45%. (2) when the time window is set as 1s-3s, the classification accuracy decreases compared with 0.5-3s. The result indicates that under unsuitable time window, some discriminative features related to motor imagery may lost.

IV. CONCLUSION

The algorithm in this paper is a combination of advantages of FBCSP and voting mechanism, which reduces the potential negative impact caused by manually band selection meanwhile simplify the calculation process. And the preliminary result shows an average accuracy of 68.6% for three-class motor imagery, which is an encouraging result in motor imagery pattern recognition. And the three-class classification strategy in this paper demonstrated that it is possible to use several binary-class classifier for multi-class motor imagery classification. The result also suggests that right-hand and left-hand motor imagery have some potential commonalities and can be considered as one class 'both-hands' in motor imagery based BCI system.

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